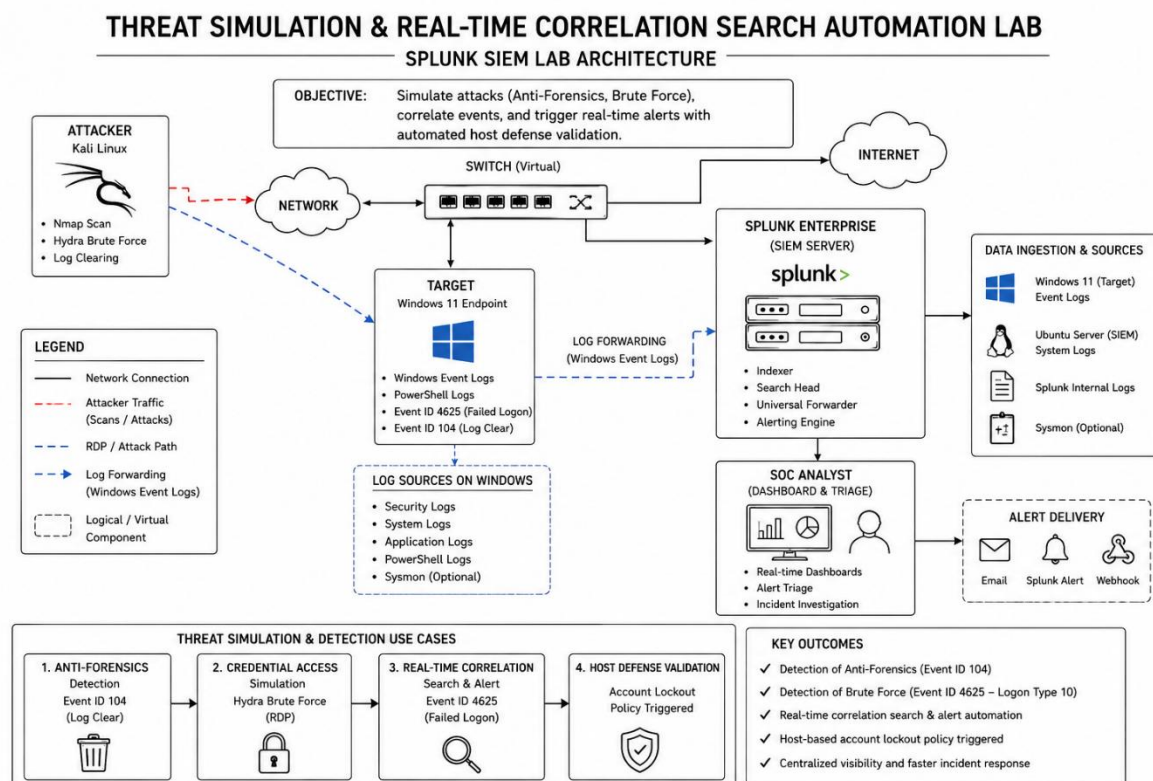


Threat Simulation & Real-Time Correlation Search Automation

Author: Qadoos Tariq Bhat

Bachelor of Computer Science (Hons.) in Cybersecurity

Objective: Threat Simulation & Real-Time Correlation Search Automation (Splunk Core Use Case Playbook)



Executive Summary

This project focused on adversary emulation and detection engineering within Splunk Enterprise. Multiple attack scenarios were executed against a Windows endpoint to generate security telemetry, which was then analyzed and transformed into actionable security alerts using custom correlation searches.

The project demonstrates practical experience in threat simulation, brute-force attack detection, anti-forensics monitoring, alert engineering, and security event correlation aligned with industry-recognized threat frameworks.

Technologies Used

- Splunk Enterprise
- Kali Linux
- Windows 11
- Hydra
- PowerShell

Frameworks & Methodologies

- MITRE ATT&CK
- Detection Engineering
- Threat Simulation
- Security Operations Center (SOC)

Skills Demonstrated

- Correlation Search Development
- Alert Engineering
- Threat Detection
- Security Monitoring
- Event Analysis
- Incident Triage
- Attack Simulation

Detection Use Cases

Credential Access

- Event ID 4625
- Hydra Brute Force Simulation
- Failed Authentication Detection

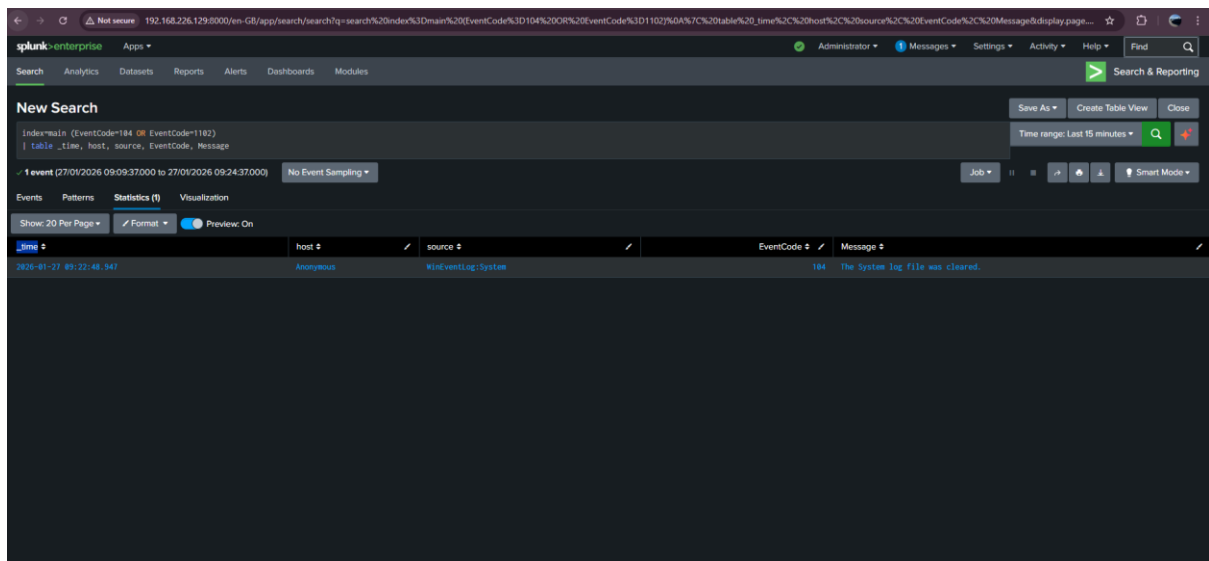
Defense Evasion

- Event ID 104
- Windows Log Clearing Detection
- Anti-Forensics Monitoring

Key Achievements

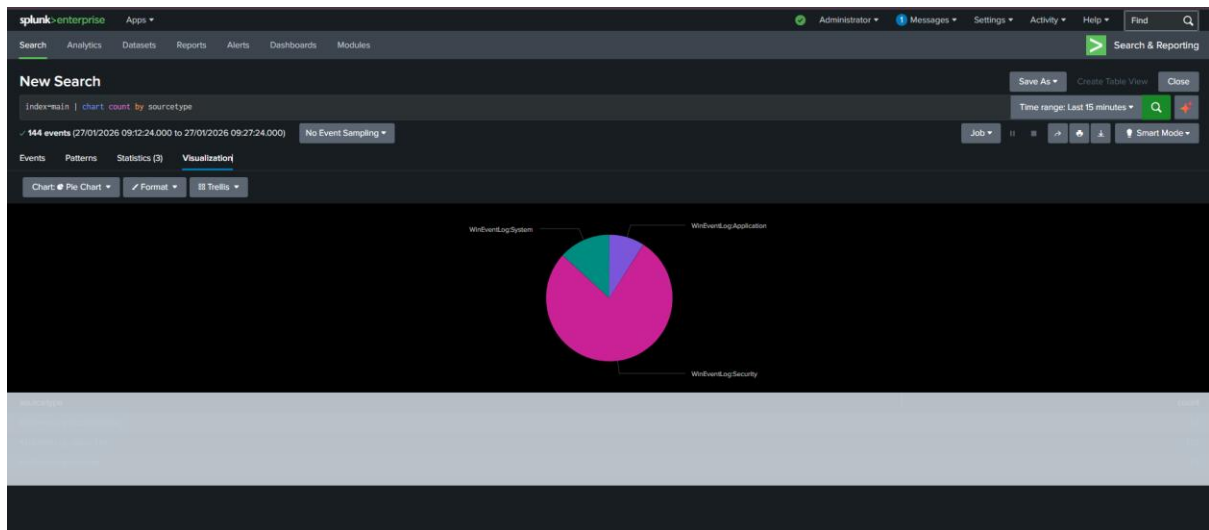
- Built custom correlation searches
- Developed real-time alerting logic
- Simulated brute-force attacks
- Detected anti-forensics activity
- Validated host-based defensive controls

Anti-Forensics Detection & Immutability Validation



Simulating and Detecting Log Tampering: I manually cleared the Windows Event Logs to simulate an anti-forensics technique used by adversaries to hide their activity. Splunk successfully captured Event ID 104, allowing for immediate detection of the 'Log Clear' event. This demonstrates the critical value of off-site log aggregation for maintaining an immutable audit trail.

Logging Health & Data Ingestion Diversity



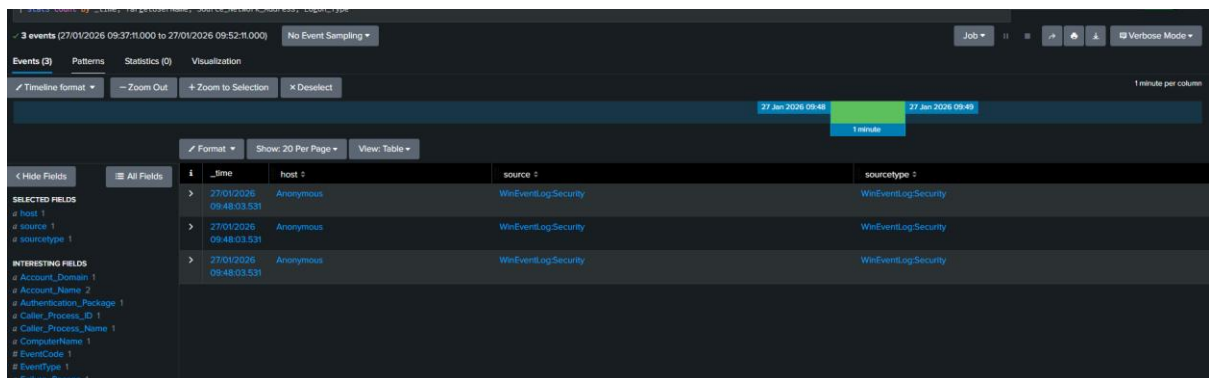
Operational Telemetry Overview: A visualization of ingested data sources within the SOC lab. This chart represents the distribution of telemetry across the Application, Security, and System channels, providing an at-a-glance view of the environment's logging health and data diversity.

Adversarial Emulation & Credential Access Simulation

```
kali@kali: ~  
└─$ hydra -i Administrator0 -p 'rongpass1,rongpass2,rongpass3' 192.168.226.128 rdp  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-01-27 04:47:52  
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3 to wait between connection to allow the server to recover  
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)  
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.  
[WARNING] Restorefile (you have 10 seconds to abort... (use option -i to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore  
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:l/p:l), -1 try per task  
[DATA] attacking rdp://192.168.226.128:3389/  
1 of 1 target completed, 0 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-01-27 04:48:03  
  
kali@kali: ~  
└─$
```

Using Hydra on Kali Linux to perform a targeted RDP brute-force attack against the Windows endpoint (192.168.226.128). This simulates the 'Credential Access' tactic from the MITRE ATT&CK framework.

Telemetry Extraction & Attack Attribution Parsing

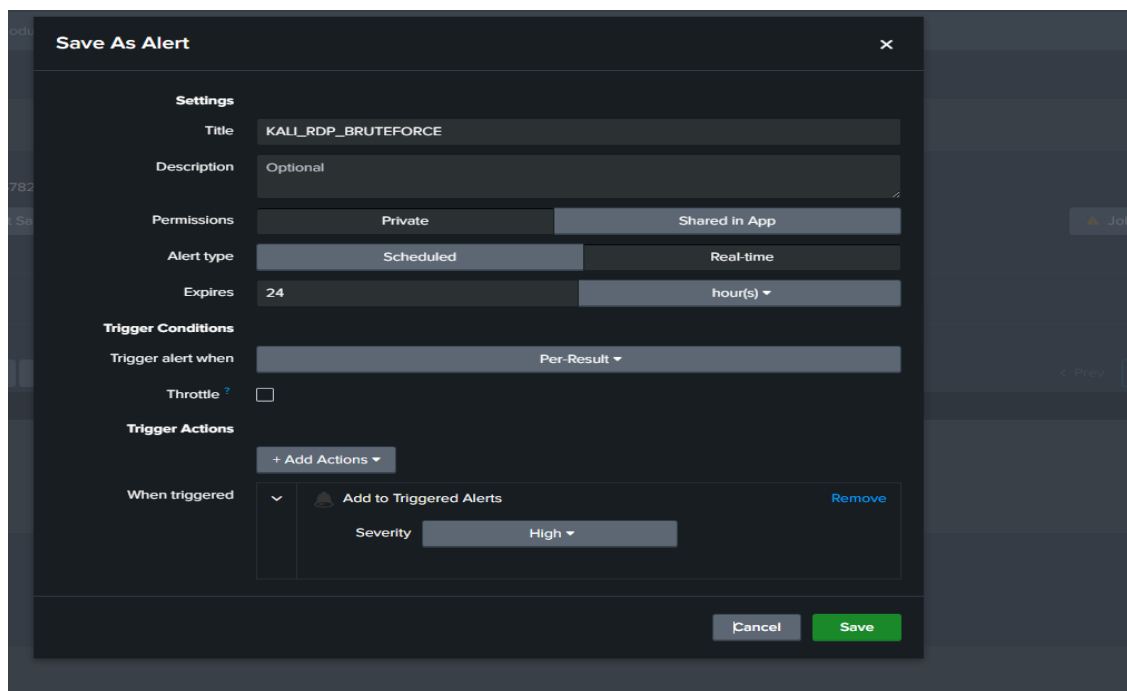


The screenshot displays the Splunk search interface with a table of event data. The search criteria at the top are '3 events (27/01/2026 09:37:11.000 to 27/01/2026 09:52:11.000)' and 'No Event Sampling'. The table has columns for _time, host, source, and sourcetype. The data shows four events from 27/01/2026 09:48:03.531 to 09:48:03.531, all from host 'Anonymous' and source 'WinEventLogSecurity', with sourcetype 'WinEventLogSecurity'. The left sidebar shows 'SELECTED FIELDS' (host, source, sourcetype) and 'INTERESTING FIELDS' (Account_Domain, Account_Name, Authentication_Package, Caller_Process_ID, Caller_Process_Name, ComputerName, EventCode, EventType, Failure_Reason).

_time	host	source	sourcetype
27/01/2026 09:48:03.531	Anonymous	WinEventLogSecurity	WinEventLogSecurity
27/01/2026 09:48:03.531	Anonymous	WinEventLogSecurity	WinEventLogSecurity
27/01/2026 09:48:03.531	Anonymous	WinEventLogSecurity	WinEventLogSecurity
27/01/2026 09:48:03.531	Anonymous	WinEventLogSecurity	WinEventLogSecurity

Splunk successfully ingested the Event ID 4625 (Failed Logon) from the Windows target. The telemetry confirms the source IP as the Kali attacker and identifies 'Logon Type 10', which is the specific signature for a Remote Desktop (RDP) authentication attempt.

Real-Time Correlation Search & Alert Automation



The screenshot shows the 'Save As Alert' configuration window. The 'Title' is 'KALI_RDP_BRUTEFORCE'. The 'Description' is 'Optional'. The 'Permissions' are set to 'Private'. The 'Alert type' is 'Real-time'. The 'Expires' is '24 hour(s)'. The 'Trigger Conditions' are set to 'Per-Result'. The 'Throttle' checkbox is unchecked. The 'Trigger Actions' section shows '+ Add Actions'. The 'When triggered' dropdown is set to 'Add to Triggered Alerts' with a 'Severity' of 'High'. The 'Cancel' and 'Save' buttons are at the bottom right.

Settings
Title: KALI_RDP_BRUTEFORCE
Description: Optional
Permissions: Private
Alert type: Real-time
Expires: 24 hour(s)
Trigger Conditions: Per-Result
Throttle: ☐
Trigger Actions: + Add Actions
When triggered: Add to Triggered Alerts (Severity: High)

Configuring a Real-time Correlation Search. By setting the alert to 'Per-Result' for Event ID 4625 originating from the Kali subnet, I have automated the detection of RDP brute-force attempts. This ensures that the SOC is notified the moment the threshold for unauthorized access attempts is crossed.

Defense-in-Depth Control Validation & Host Mitigation

```
Session Actions Edit View Help
(kali@kali)-[~]
$ hydra -l Administrator0 -p 'password123,admin,123456,Password1,qwerty' -t 1 -V 192.168.226.128 rdp
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** i
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-01-29 10:56:54
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), -1 try per task
[DATA] attacking rdp://192.168.226.128:3389/
[ATTEMPT] target 192.168.226.128 - login "Administrator0" - pass "password123,admin,123456,Password1,qwerty" - 1 of 1 [child 0] (0/0)
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-01-29 10:56:57
```

During the RDP brute-force simulation, the Windows target's account lockout policy was triggered. This automated defensive response successfully mitigated the attack by disabling the targeted account after a threshold of failed attempts was reached, effectively blocking further adversarial progression.

Conclusion

The project successfully validated Splunk's ability to detect and correlate suspicious activities across multiple attack scenarios. By transforming raw event data into actionable alerts, the environment demonstrated how SIEM platforms can support rapid detection and incident response workflows.

This project enhanced skills in attack simulation, detection engineering, correlation rule development, security monitoring, and threat detection operations.